

Introduction to conjoint analysis



Welcome!

Housekeeping!

- Be kind and curious!
- Slack and Zoom chat
- Ask questions

Schedule

Day 1

Time	Activity
10:30–12:30	Welcome + Intro to conjoint analysis
12:30–13:30	<i>Break</i>
13:30–15:00	Designing conjoint experiments

Day 2

Time	Activity
10:30–12:30	Conjoint data and causal inference (part 1)
12:30–13:30	<i>Break</i>
13:30–15:00	Conjoint data and causal inference (part 2)

Schedule

Day 3

Time	Activity
10:30–12:30	Conjoint data and respondent preferences (part 1)
12:30–13:30	<i>Break</i>
13:30–15:00	Conjoint data and respondent preferences (part 2)

Day 4

Time	Activity
10:30–12:30	Implementing and fielding conjoint surveys (part 1)
12:30–13:30	<i>Break</i>
13:30–15:00	Implementing and fielding conjoint surveys (part 2)

About me

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- Assistant professor of ~~public policy~~ international relations,
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- Data visualization, statistics, and causal inference



Meeting you where you are

This course is designed for someone who:

- Is familiar with R (and ideally tidyverse)
- Has some experience with statistical modeling with linear models
- Maybe has an idea for a conjoint experiment

You'll learn:

- What conjoint analysis is
- How to measure causal effects and explore “product” and “consumer” preferences
- How to design and run your own conjoint experiments

Course structure

My turn

- Lecture segments
- Feel free to just watch, take notes, browse docs, or tinker around with the code
- Lots of live-coding! Most content lives in Quarto files

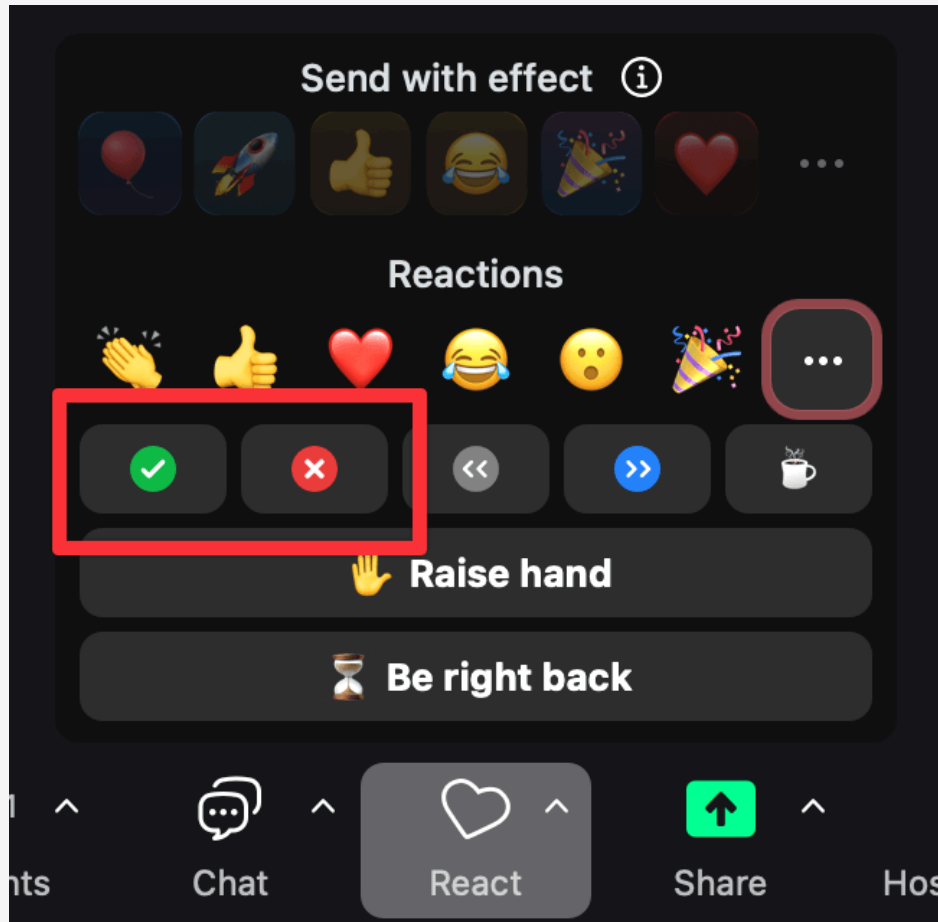
Your turn

- Exercises for you to do
- Work on your own or with others

Bring your own ideas!

- The best way to learn is by building something
- I've given you playground code, but it's fairly basic. *This is by design!*
- You'll have time during the “Your turn”s to to play around and experiment. Try to use these new principles and skills to create something!

Getting help



Use Zoom reactions

-  =
“I’m stuck and need help!”
-  =
“I finished the exercise”

Ask longer, more detailed questions
in Slack

Your turn

Introduce yourself:

- Name and professional affiliation
- On a scale of 1–10, how well do you know...
 - R?
 - Tidyverse?
 - Statistical models like OLS, multinomial logit, and Bayesian methods?
 - Conjoint analysis?
- What do you hope to get out of this course?

04:00

What is conjoint analysis?

The problem of preferences

How do you measure what people *really* want?

- **Direct questions:** “Do you care about the environment?” → Everyone says yes
- **Stated vs. revealed preference:** What people say $\neq$ what they do
- **Single attributes:** “Is healthcare important?” Of course! But *how* important vs. cost? Do they interact?
- **Ranking:** People rank things differently depending on context

People make trade-offs. Survey experiments that don't force trade-offs can give misleading answers.

Ice cream!

It's hot outside and you want to buy some ice cream. You go to your local ice cream shop and find a menu with all these possibilities:

- Vanilla, chocolate, or strawberry
- Served in a bowl or served in a waffle cone
- No extra toppings, topped with nuts, or topped with sprinkles
- Priced at \$3, \$4, or \$5

Which kind do you want? Why?

No ice cream maximizes everyone's happiness.

There are too many trade-offs

What is conjoint analysis?

Conjoint analysis is a family of survey-based techniques where respondents evaluate or choose among hypothetical profiles described by multiple attributes, revealing their preferences through the trade-offs they make.

conjoint = *considered jointly*

Respondents evaluate attributes **together** (as bundles), not separately

Use statistics to uncover preferences from these bundles

andhs.co/icecream

Where did it come from?

Psychology (1960s)

Luce, R. Duncan, and John W. Tukey. 1964. "Simultaneous Conjoint Measurement: A New Type of Fundamental Measurement." *Journal of Mathematical Psychology* 1 (1): 1–27. [https://doi.org/10.1016/0022-2496\(64\)90015-X](https://doi.org/10.1016/0022-2496(64)90015-X).

Originally developed to measure preferences of sets of features

JOURNAL OF MATHEMATICAL PSYCHOLOGY: 1, 1-27 (1964)

Simultaneous Conjoint Measurement: A New Type of Fundamental Measurement

R. DUNCAN LUCE^{1, 2}

University of Pennsylvania, Philadelphia, Pennsylvania

AND

JOHN W. TUKEY^{1, 3}

*Princeton University, Princeton, New Jersey, and Bell Telephone Laboratories,
Murray Hill, New Jersey*

The essential character of what is classically considered, e.g., by N. R. Campbell, the fundamental measurement of extensive quantities is described by an axiomatization for the comparison of effects of (or responses to) arbitrary *combinations* of "quantities" of a *single specified kind*. For example, the effect of placing one arbitrary combination of masses on a pan of a beam balance is compared with another arbitrary combination on the other pan. Measurement on a ratio scale follows from such axioms. In this paper, the essential character of simultaneous *conjoint* measurement is described by an axiomatization for the comparison of effects of (or responses to) *pairs* formed from *two specified kinds* of "quantities." The axioms apply when, for example, the effect of a pair consisting of one mass and one difference in gravitational potential on a device that responds to momentum is compared with the effect of another such pair. Measurement on interval scales which have a common unit follows from these axioms; usually these scales can be converted in a natural way into ratio scales.

A close relation exists between conjoint measurement and the establishment of response measures in a two-way table, or other analysis-of-variance situations, for which the "effects of columns" and the "effects of rows" are additive. Indeed, the discovery of such measures, which are well known to have important practical advantages, may be viewed as the discovery, via conjoint measurement, of fundamental measures of the row and column variables. From this point of view it is natural to regard conjoint measurement as factorial measurement.

¹ The hospitality of the Center for Advanced Study in the Behavioral Sciences and of Stanford University is gratefully acknowledged. We wish to thank Francis W. Irwin and Richard Robinson for careful readings of an earlier version which have led to corrections and clarifications of the argument.

² Research supported in part by the National Science Foundation grant NSF G-17637 to the University of Pennsylvania.

³ Research in part at Princeton University under the sponsorship of the Army Research Office (Durham).

Marketing research (1970s)

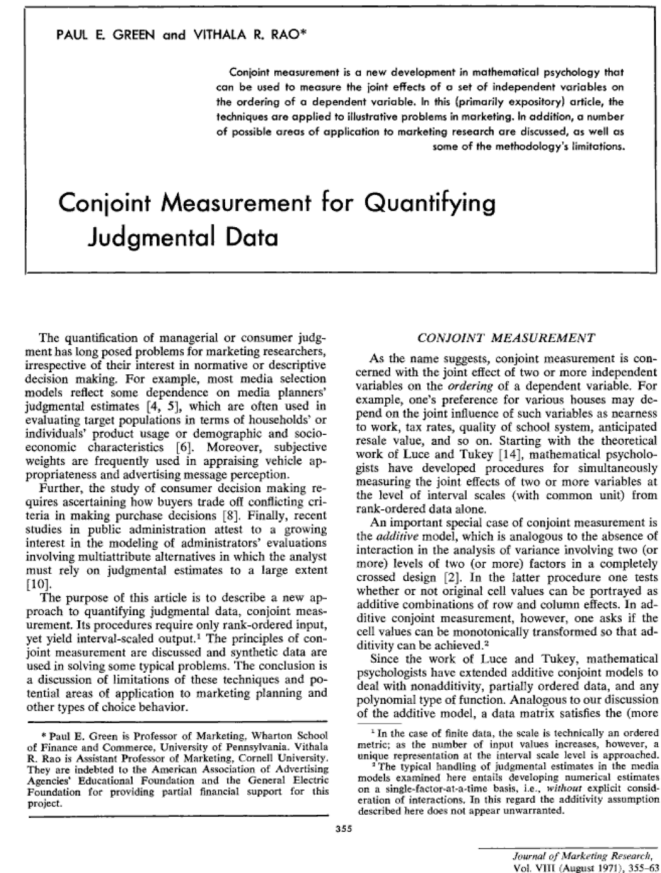
Green, Paul E., and Vithala R. Rao. 1971. "Conjoint Measurement for Quantifying Judgmental Data." *Journal of Marketing Research* 8 (3): 355.

<https://doi.org/10.2307/3149575>.

Adapted to answer marketing questions:

- How should we price our product?
- Which features should we invest in?
- What product will maximize market share?

Uses: consumer electronics, airlines, credit cards, pharmaceuticals, automobiles



Apple vs. Samsung (2012)

- Apple sued Samsung for patent infringement: copied features like sliding to unlock, search, autocorrect, pinch-to-zoom, etc.
- Apple hired MIT marketing professor **John Hauser** to measure value of those features
- Conjoint experiment analyzed with a hierarchical Bayesian model
- Consumers would be willing to spend between \$32–\$102 extra for these features
- These results (and other bits of evidence) scaled up to \$2 billion in damages awarded to Apple

Causal inference (2010s)

Hainmueller, Jens, Daniel J. Hopkins, and Teppei Yamamoto. 2014. "Causal Inference in Conjoint Analysis: Understanding Multidimensional Choices via Stated Preference Experiments." *Political Analysis* 22 (1): 1–30.

<https://doi.org/10.1093/pan/mpt024>.

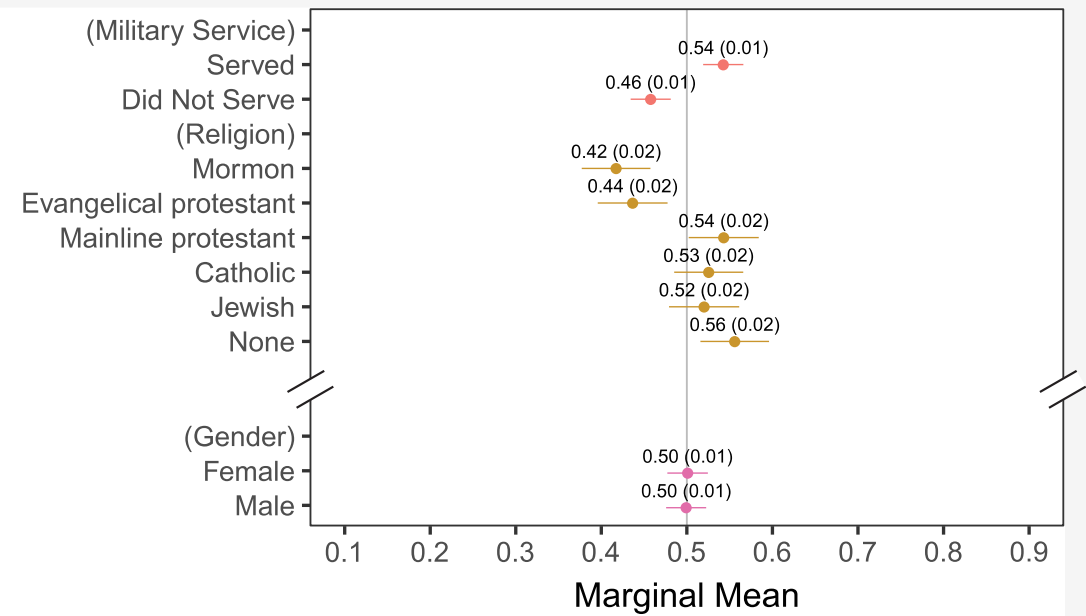
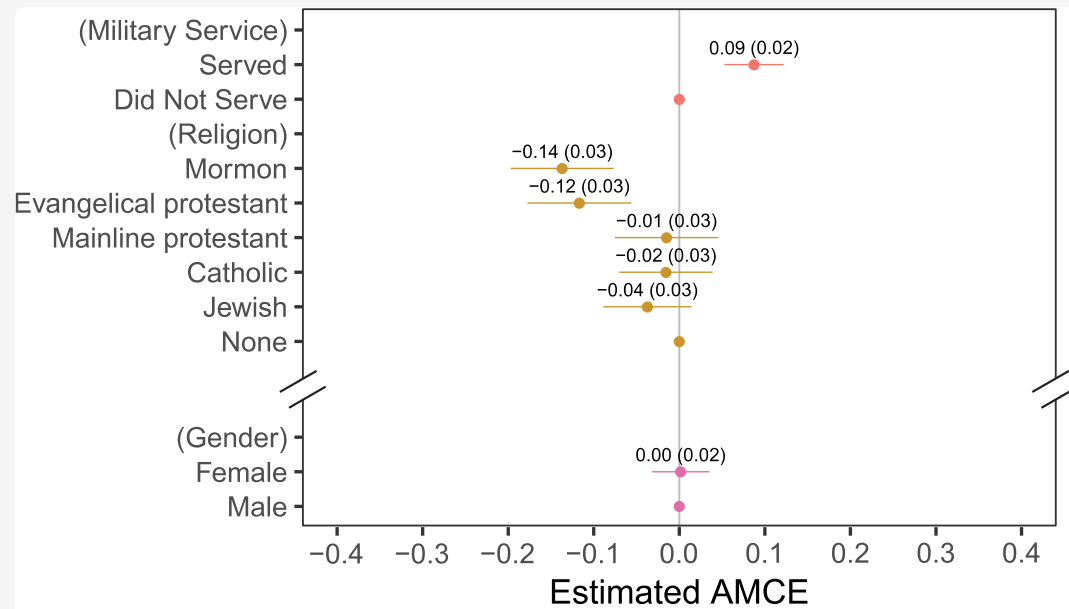
Instead of analyzing a whole constellation of attributes, look at specific levers individually.

Conjoints are like RCTs, just fancier.

Since treatment assignment is random, we can find the effect on the selection of probability (or favorability) when changing an attribute.

Uses: Measure effects of political candidate characteristics, issue salience, organization characteristics





**What can we estimate
with it?**

Preferences

Marketing

- How much do people value the overall constellation of features?
- What product will sell best?
- What are people willing to pay for this feature?
- Estimands: **utilities, attribute importance, willingness to pay, market shares and simulations**

Causal inference

Social science

- Does feature X make people more or less likely to choose a “product”?
- What is the average causal effect of a specific feature?
- Estimands: **average marginal component effect (AMCE), marginal means, average feature choice probability (AFCP)**

Same data and same models,
completely different estimands

Crossing traditions!

Check for updates

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Special Issue Article

Who Cares about Crackdowns? Exploring the Role of Trust in Individual Philanthropy

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Abstract

The phenomenon of closing civic space has adversely impacted international non-governmental organization (INGO) funding. We argue that individual private donors can be important in sustaining the operations of INGOs working in repressive contexts. Individual donors do not use the same performance-based metrics as official aid donors. Rather, trust can be an important component of individual donor support for nonprofits working towards difficult goals. How does trust in charitable organizations influence individuals' preferences to donate, especially when these groups face crackdown? Using a simulated market for philanthropic donations based on data from a nationally representative sample of individuals in the United States who regularly donate to charity, we find that trust in INGOs matters substantially in shaping donor preferences. Donor profiles with high levels of social trust are likely to donate to INGOs with friendly relationships with host governments. This support holds steady if INGOs face criticism or crackdown. In contrast, donor profiles with lower levels of social trust prefer to donate to organizations that do not face criticism or crackdown abroad. The global crackdown on NGOs may thus possibly sour NGOs' least trusting individual donors. Our findings have practical implications for INGOs raising funds from individuals amid closing civic space.

Policy implications

- In the wake of the global crackdown on civil society, international NGOs (INGOs) need to maintain access to multiple sources of funding, including individual donors.
- It is critical for INGOs to understand what motivates individuals to donate, especially if INGOs face legal crackdown in their host countries.
- Trust in political institutions and in charitable organizations matters substantially in shaping donor preferences.
- We find that potential donors with low social trust tend to be wary of negative host-country relationships and prefer to donate to INGOs that are friendly with their host governments, while potential donors with high social trust are more willing to stick with legally besieged INGOs.
- Donors that are the most trusting and involved may remain supportive when charities face government criticism and crackdown. The global crackdown on NGOs may also possibly sour NGOs' least trusting individual donors.

1. Who cares about crackdowns? Exploring the role of trust in individual philanthropy

In 2016, Human Rights Watch claimed that civil society was under more aggressive attack than at any time in recent memory (Roth, 2016). Globally, governments are repressing civil society organizations (CSOs) and non-governmental organizations (NGOs) – a phenomenon known as 'closing civic space' (Carothers and Brechenmacher, 2014; CIVICUS, 2017; Dupuy et al., 2015, 2016).¹ Legal restrictions, or what we refer to as legal crackdowns, are a core part of these efforts. These crackdowns create barriers to entry, funding, and advocacy for NGOs in an effort to control, obstruct, and repress these organizations. Barriers to funding are especially pervasive and restrict the ability of NGOs to secure financial resources. Restrictive states may prevent the transfer of foreign funds to NGOs based on the origin and purposes of these funds. Due to these laws, official foreign aid channeled through international NGOs (INGOs) has decreased in repressive countries (Brechenmacher, 2017; Chaudhry and Heiss, 2018; Dupuy and Prakash, 2018). However, philanthropy from private donors and foundations is not as adversely affected (McGill, 2018). Many foundations have continued channeling

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Check for updates

Article

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Navigating Hostility:
The Effect of Nonprofit
Transparency and
Accountability on Donor
Preferences in the Face of
Shrinking Civic Space

Suparna Chaudhry¹ , Marc Dotson² ,
and Andrew Heiss³ 

Abstract

Governments across the world have increasingly used laws to restrict the work of nonprofits, which has led to a reduction in public or official foreign aid directed toward these groups. Many international nonprofits, in response, have turned to individual donors to offset the loss of traditional funding. What are individual donors' preferences regarding donating to legally besieged nonprofits abroad? We conducted a conjoint experiment on a nationally representative sample of likely donors in the United States and found that learning about host government criticism and legal restrictions on nonprofits decreases individuals' preferences to donate to them. However, organizational features such as financial transparency and accountability can protect against this dampening effect. Our results have important implications both for understanding private international philanthropy and how nonprofits can better frame their fundraising appeals at a time when they are facing restrictive civic spaces and hostile governments abroad.

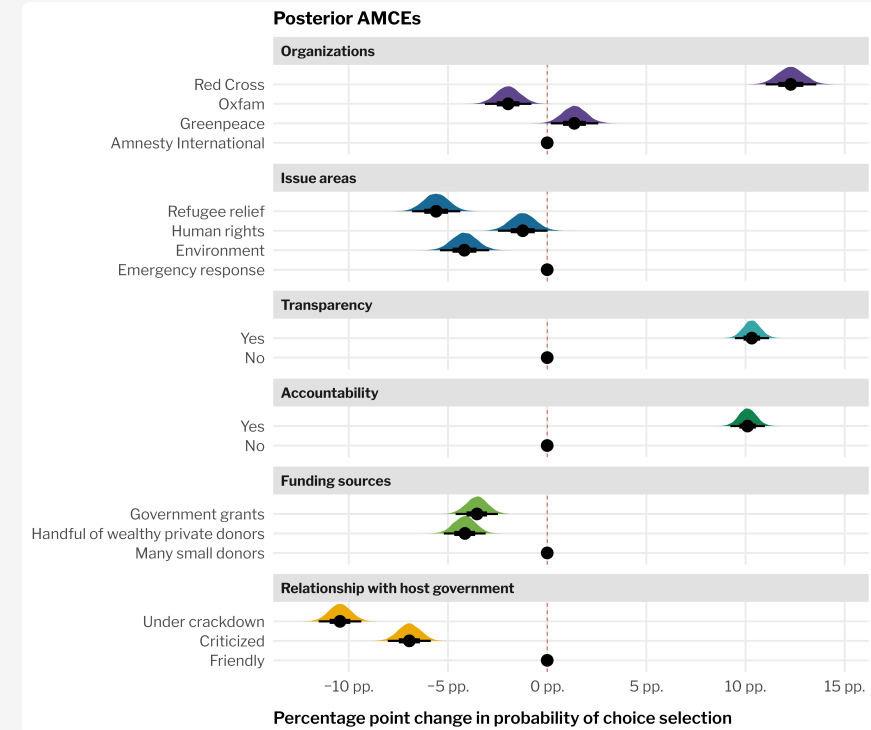
Keywords

philanthropy, conjoint experiments, donor heuristics, repression, NGOs, civil society, nonprofits

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Crossing traditions!



Chaudhry, Suparna, Marc Dotson, and Andrew Heiss. 2021. "Who Cares about Crackdowns? Exploring the Role of Trust in Individual Philanthropy." *Global Policy* 12 (S5): 45–58. <https://doi.org/10.1111/1758-5899.12984>.

Chaudhry, Suparna, Marc Dotson, and Andrew Heiss. 2025. "Navigating Hostility: The Effect of Nonprofit Transparency and Accountability on Donor Preferences in the Face of Shrinking Civic Space." *Nonprofit and Voluntary Sector Quarterly*, 1–27. <https://doi.org/10.1177/08997640251348654>.